

2 Carnot cycles and Entropy

The problems are roughly in order of difficulty. Those indicated with ♡ are those which it is most important to work on. Any labelled with ♣ are the hardest ones, which might only occur as a "sting in the tail" at the end of a long examination question.

Problem 2.1 A Carnot cycle

Define:

$Q_2 = 2000 \text{ J}$ = heat intake at hot reservoir, T_2 .

Q_1 = heat output at cold reservoir T_1 .

W = work performed.

(i) Since this is a Carnot engine,

$$\frac{Q_1}{Q_2} = \frac{T_1}{T_2} \text{ so } Q_1 = \frac{T_1}{T_2} Q_2 = \frac{350}{500} \times 2000 \text{ J} = 1400 \text{ J} .$$

(ii) By conservation of energy, $W = Q_2 - Q_1 = 600 \text{ J}$.

(iii)

$$\eta = 1 - \frac{Q_1}{Q_2} = 1 - \frac{T_1}{T_2} = 30\% .$$

(iv) There is no change in entropy on the (isentropic) adiabatic changes

$$\text{At the hot reservoir, } \Delta S_{\text{hot}} = \frac{Q_2}{T_2} = \frac{2000}{500} = +4 \text{ JK}^{-1}$$

$$\text{At the cold reservoir, } \Delta S_{\text{cold}} = \frac{Q_1}{T_1} = -\frac{1400}{350} = -4 \text{ JK}^{-1} .$$

Problem 2.2 Second law – equivalence of Kelvin and Clausius

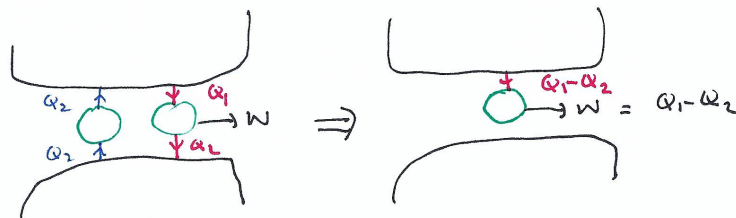


Figure 2.1: Left: An impossible Clausius heat pump and a Carnot heat engine. Right: The composite – note cancellation of the heat flow with the cold reservoir – is a Kelvin-impossible engine.

Let's show that a machine that violates Clausius' statement will also violate Kelvin's statement. Starting from a heat pump extracting heat Q_2 in one cycle from a cold reservoir to

a hot reservoir, we want to build an engine that produces work from heat extracted from the hot reservoir only. Let's therefore consider an engine extracting heat Q_1 from the hot reservoir and producing the work $W = Q_1 - Q_2$ by rejecting heat Q_2 to the cold reservoir. Of course, we have to choose W such that the efficiency of the engine is less than or equal to the Carnot efficiency. If we now consider the heat pump and the engine, the combined system does not exchange heat with the cold reservoir (the two contributions cancel), and at every cycle of the system transforms the heat $Q_1 - Q_2$ into work, which is impossible according to Kelvin's statement.

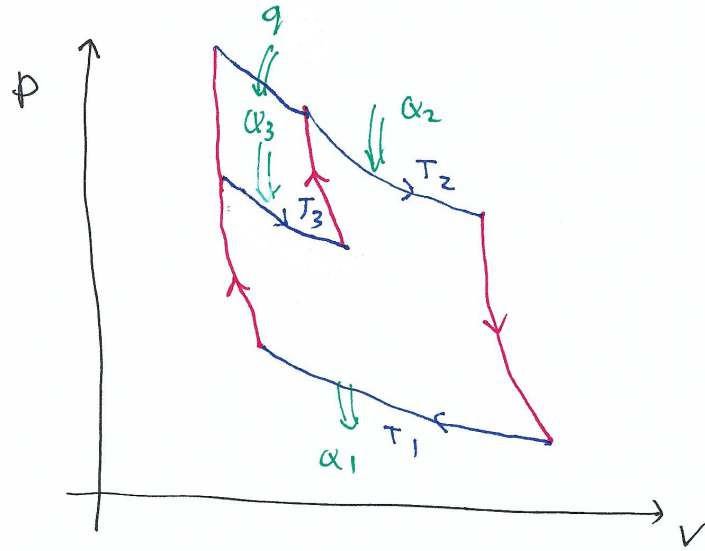


Figure 2.2: Three-reservoir cycle including the extra section (with heat flow q), to create a comparator two-reservoir cycle.

♣ Problem 2.3 Three reservoirs do not improve efficiency

Consider Fig. (2.2). Firstly consider the efficiency of the three-reservoir cycle, defined as usual by the ratio of work to heat input from the hot reservoir(s). Let η_3 be the efficiency of the three-reservoir cycle:

$$\eta_3 = \frac{Q_2 + Q_3 - Q_1}{Q_2 + Q_3} = 1 - \frac{Q_1}{Q_2 + Q_3} . \quad (2.1)$$

But we may add and subtract the heat for the section which would make up the two-reservoir case, q , to obtain:

$$\eta_3 = 1 - \frac{Q_1}{Q_2 + q - (q - Q_3)} < 1 - \frac{Q_1}{Q_2 + q} = \eta_2 , \quad (2.2)$$

where η_2 is the efficiency of the two-temperature cycle including the extension of the upper temperature isotherm to include q . Note that $q - Q_3 > 0$ as the mini-Carnot formed from those isotherms has a positive area – i.e. work performed. Thus the denominator for the three-temperature expression is smaller than that for the two-temperature case, so *decreases* efficiency.

Problem 2.4 Integrating Entropy

Start by rearranging to obtain

$$dS = \frac{p dV}{T} + \frac{dU}{T} . \quad (2.3)$$

For an ideal gas, $p/T = nR/V$ where, in this case, $n = 1$ and

$$dU = C_v dT .$$

So in Eq. (2.3),

$$dS = \frac{R dV}{V} + \frac{C_v dT}{T} . \quad (2.4)$$

Since $S(V, T)$ is a function of state, we may choose any path between the initial, (V_0, T_0) , and final, (V, T) , states. For example initially keep T fixed, so $dT = 0$, and integrate both sides of Eq. (2.4) to obtain

$$S(V, T_0) - S(V_0, T_0) = \int_{V_0}^V \frac{R}{V'} dV' = R \ln(V/V_0) . \quad (2.5)$$

Then integrate Eq. (2.4) with respect to T from (V, T_0) to (V, T) , keeping V constant (i.e. $dV = 0$) to find

$$S(V, T) - S(V, T_0) = C_v \int_{T_0}^T \frac{dT'}{T'} = C_v \ln(T/T_0) . \quad (2.6)$$

Adding the left hand sides and right hand sides of Eq. (2.5) and Eq. (2.6), to construct the contribution from the combined path, we see

$$\begin{aligned} S(V, T) - S(V, T_0) + S(V, T_0) - S(V_0, T_0) &= R \ln(V/V_0) + C_v \ln(T/T_0) \\ \Rightarrow S(V, T) &= S(V_0, T_0) + R \ln(V/V_0) + C_v \ln(T/T_0) . \end{aligned}$$

Check that setting $T = T_0$ and $V = V_0$ yields correct identity, $S(V_0, T_0) = S(V_0, T_0)$.

Problem 2.5 Homely entropy examples

- (i) You can treat the air as a reservoir at temperature $T = 20^\circ\text{C} = 293\text{K}$, unaffected by the energy emitted by the bulb. Then:

$$\begin{aligned} \text{Energy emitted from bulb} \quad \Delta Q &= 100 \times 60 = 6000 \text{ J} \\ \text{Entropy increase of room} \quad \Delta S &= \frac{\Delta Q}{T} = \frac{6000}{293} \text{ JK}^{-1} = 20.5 \text{ JK}^{-1} . \end{aligned}$$

After the bulb has been turned off, it is back in its original state. Since entropy is a function of state, the entropy of the bulb is the same as at the beginning and the entropy change of the universe is $+20.5 \text{ JK}^{-1}$.

- (ii) This is very similar to a previous problem done in the lectures. Begin by evaluating the final temperature T_0 by demanding that the heat ΔQ flowing out of the hot water matches that flowing into the cold water. This implies

$$\Delta Q = 0.2C(80 - T_0) = 0.8C(T_0 - 10) ,$$

where C is the specific heat capacity and the temperatures are in centigrade. Solving this yields $T_0 = 24^\circ\text{C} = 297\text{ K}$.

Then the entropy change to the 0.2 litres which starts at 80°C is

$$\Delta S_{\text{hot}} = 0.2C \int_{353}^{297} \frac{dT}{T} = 0.2C \ln \frac{297}{353},$$

and similarly, the entropy change to the 0.8 litres which starts at 10°C is

$$\Delta S_{\text{cold}} = 0.8C \int_{283}^{297} \frac{dT}{T} = 0.8C \ln \frac{297}{283}.$$

The entropy change to the Universe is then ...

$$\Delta S_{\text{Univ}} = \Delta S_{\text{hot}} + \Delta S_{\text{cold}} = 4190 \left(0.2 \ln \frac{297}{353} + 0.8 \ln \frac{297}{283} \right) = 17 \text{ JK}^{-1}.$$

Problem 2.6 Water to steam

Energy to raise 1 kg of water from 0 to $100^\circ\text{C} = 4.2 \times 100 = 420 \text{ kJ}$.

Energy to evaporate at 100°C at constant pressure $= 2.26 \times 10^3 \text{ kJ}$.

Work done by the water/steam in expansion $= p\Delta V = 1.01 \times 10^5 \times 1.67 \text{ J} = 169 \text{ kJ}$.

So internal energy of steam relative to that of water $= 420 + 2260 - 169 = 2511 \text{ kJ}$

Problem 2.7 Entropy of a cooling block

Be careful to convert all temperatures to kelvins. The final temperature of the block is T_2 .

Following a reversible path from T_1 to T_2 , the change in entropy of the block is

$$\Delta S = \int_{T_1}^{T_2} \frac{dQ^{\text{rev}}}{T} = \int_{T_1}^{T_2} \frac{CdT}{T} = C \ln \frac{T_2}{T_1} \simeq -110 \text{ JK}^{-1}.$$

The block loses heat, so its entropy goes down. The reservoir receives the total heat lost by the block, $C(T_1 - T_2)$, at a constant temperature T_2 , which gives an entropy change for the reservoir of

$$\Delta S = \frac{C(T_1 - T_2)}{T_2} \simeq 124 \text{ JK}^{-1}.$$

Problem 2.8 Joule's second experiment

The free expansion involves no change in the temperature, but is irreversible. To find the ΔS that occurs during the expansion, recompress the gas *reversibly* and isothermally back to the original volume and calculate the work done. Since there is no change in internal energy (a function of the temperature only for an ideal gas), the First Law tells us that the heat generated is equal to the work done in the recompression.

First calculate the work done on the gas.

$$W = - \int_{V_1}^{V_2} p dV = -nRT \int_{V_1}^{V_2} \frac{dV}{V} = p_1 V_1 \ln \left(\frac{V_1}{V_2} \right) = 2 \times 10^5 \times 0.2 \ln 2 \text{ J} = 2.77 \times 10^4 \text{ J},$$

So:

$$\begin{aligned} Q_{\text{out}} &= 2.77 \times 10^4 \text{ J}, \\ \Rightarrow \Delta S_{\text{reversible}} &= \frac{Q}{T} = \frac{2.77 \times 10^4}{(273 + 25)} = 93 \text{ JK}^{-1}. \end{aligned}$$

The gas has now been restored to its original state so the entropy (a function of state) must be the same as before the expansion.

Problem 2.9 Maximum work out of a hot block

The engine will operate until the temperature of the block reaches the temperature of the reservoir, T_1 . Maximum work is achieved by a Carnot engine, as it has the largest possible efficiency. Over a single cycle of this engine, the heat $\delta Q = -C dT$ (noting $dT < 0$) is extracted from the block at temperature T , producing the work with the Carnot efficiency

$$\delta W = \delta Q \eta_{\text{Carnot}} = - \left(1 - \frac{T_1}{T} \right) C dT .$$

The total work is therefore

$$W = \int_{T_2}^{T_1} \left(\frac{T_1}{T} - 1 \right) C dT = C \left(T_1 \ln \frac{T_1}{T_2} + T_2 - T_1 \right) \simeq 20 \text{ kJ} .$$

The total heat extracted from the block is $Q = C(T_2 - T_1) \simeq 90 \text{ kJ}$, for a total efficiency of $\eta = W/Q \simeq 0.22$.

Problem 2.10 T-S diagram

The Carnot cycle is formed of two isentropes (constant entropy) and two isotherms (constant temperature), therefore the cycle looks like a rectangle in the T - S plane. The orientation of the cycle is found by remembering that heat is extracted from the hot reservoir, therefore increasing the entropy on the high-temperature isotherm.

There is no heat being exchanged on the isentropes (reversible adiabatic process), and the heat transferred to the system on the reversible isotherms is, from the definition of the entropy,

$$Q_{\text{in}} = T \Delta S .$$

This means that the total heat received by the engine is the area of the cycle in the T - S plane. Over a cycle the energy is unchanged:

$$0 = \Delta U = Q_{\text{in}} + W_{\text{on}} = Q_{\text{in}} - W_{\text{by}} ,$$

therefore $W_{\text{by}} = Q_{\text{in}}$. The work done by the engine is also the area of the cycle. For any cycle, the second law says that $\delta Q_{\text{in}} \leq T_{\text{ext}} dS$ at all points of the cycle, with a strict inequality in some places for an irreversible cycle. Therefore, W_{by} , is always equal to $Q_{\text{in}} = \int \delta Q_{\text{in}}$, is less than or equal to the area of the cycle in the T_{ext} - S plane, with equality only for a reversible cycle.

Problem 2.11 Entropy of a p - T change

Let's decompose the transformation into a reversible isobaric transformation from T_1 to T_2 at pressure p_1 , followed by a reversible isothermal transformation from p_1 to p_2 at temperature T_2 . The entropy change during the isobaric transformation is

$$\Delta S = \int_{T_1}^{T_2} \frac{n c_p dT}{T} = n c_p \ln \frac{T_2}{T_1} \simeq 3.9 \text{ JK}^{-1} ,$$

with $n = 1 \text{ mol}$.

For an isothermal transformation of an ideal gas, one has $0 = dU = dQ + dW$, therefore $dQ^{\text{rev}} = -dW^{\text{rev}} = p dV$. The entropy change of the reversible transformation is

$$\Delta S = \int_{V_1}^{V_2} \frac{nR dV}{V} = nR \ln \frac{V_2}{V_1} = nR \ln \frac{p_1}{p_2} \simeq -3.4 \text{ JK}^{-1},$$

using $pV = nRT$. The total entropy change is the sum: $\Delta S \simeq 0.5 \text{ JK}^{-1}$.

It is also possible to start from the solution of the $V - T$ transformation (lecture notes, or problem (2.4)) and use $pV = nRT$ and $c_p = c_v + R$.

♣ Problem 2.12 Two-phase cycle

(i) Irrespective of the details of the design, the efficiency of the Carnot engine is

$$\eta = 1 - \frac{T_1}{T_2} = \frac{T_2 - T_1}{T_2}.$$

(ii) Since the vapour pressure is determined only by the temperature, the two isothermal parts of the Carnot cycle ($1 \rightarrow 2$) and ($3 \rightarrow 4$) are also at constant pressure. Hence Fig. (2.3) represents something like that below. .

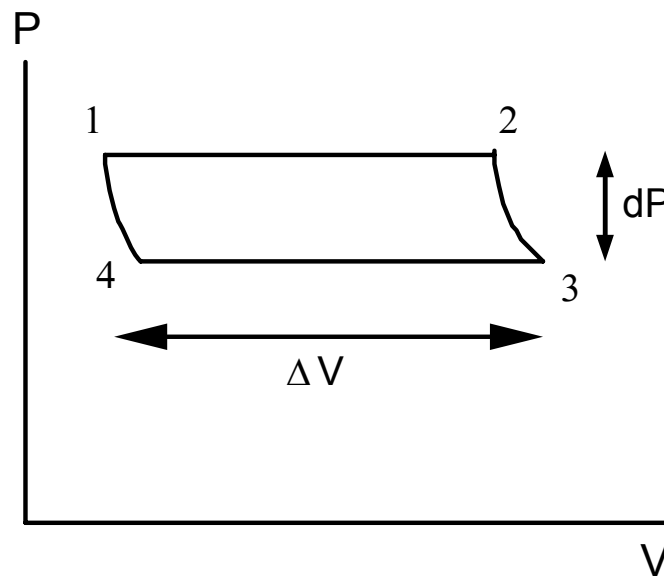


Figure 2.3: .

(iii) The net work done by the gas is the area enclosed by the loop, $\Delta V dp$.

The heat input to the system Q_2 in section ($1 \rightarrow 2$) is also the latent heat L . One expression for the efficiency is thus

$$\eta = \frac{W}{Q_2} = \frac{\Delta V dp}{L}.$$

For the small change in temperature as explained in the question, the efficiency as given in part (i) can be written as

$$\eta = \frac{T_2 - T_1}{T_2} = \frac{dT}{T} .$$

Equating the two expressions for the efficiency:

$$\frac{dT}{T} = \frac{\Delta V dp}{L} \quad \text{and hence} \quad \frac{dp}{dT} = \frac{L}{T \Delta V} .$$